

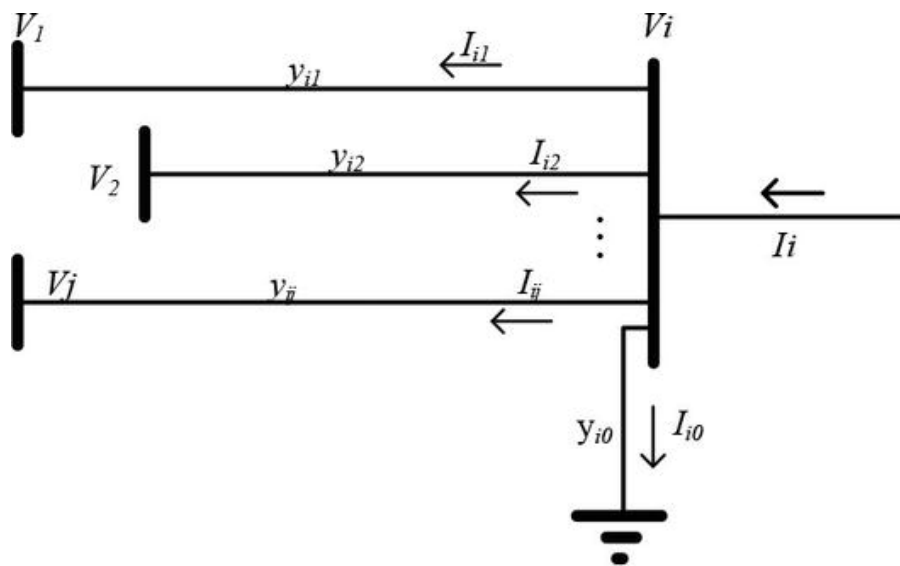
ETAP Power System Analysis and Load Flow

Table of Contents

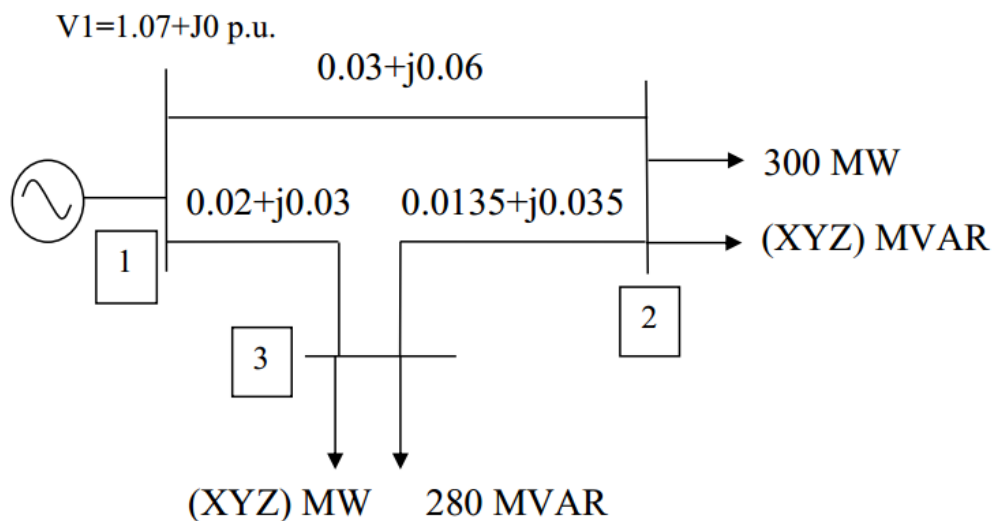
1	Introduction	2
2	Power system model:	2
3	ETAP Simulation of the power system	3
4	Snapshots of the ETAP	3
4.1	Voltages at buses 2 and 3	3
4.2	All active and reactive line losses.	4
4.3	All the currents in the lines	4
4.4	The effect of 20 MVAR capacitor bank at bus 2 on the line losses, with voltage rating same bus voltage 5	
4.5	The effect of 20 MVAR capacitor bank at bus 2 on the line losses, with 1kv rating bus voltage 5	
4.6	The effect of the voltage magnitude and angle on the active and reactive power flow	6
4.6.1	Case 1:	7
4.6.2	Case 2:	7

1 Introduction

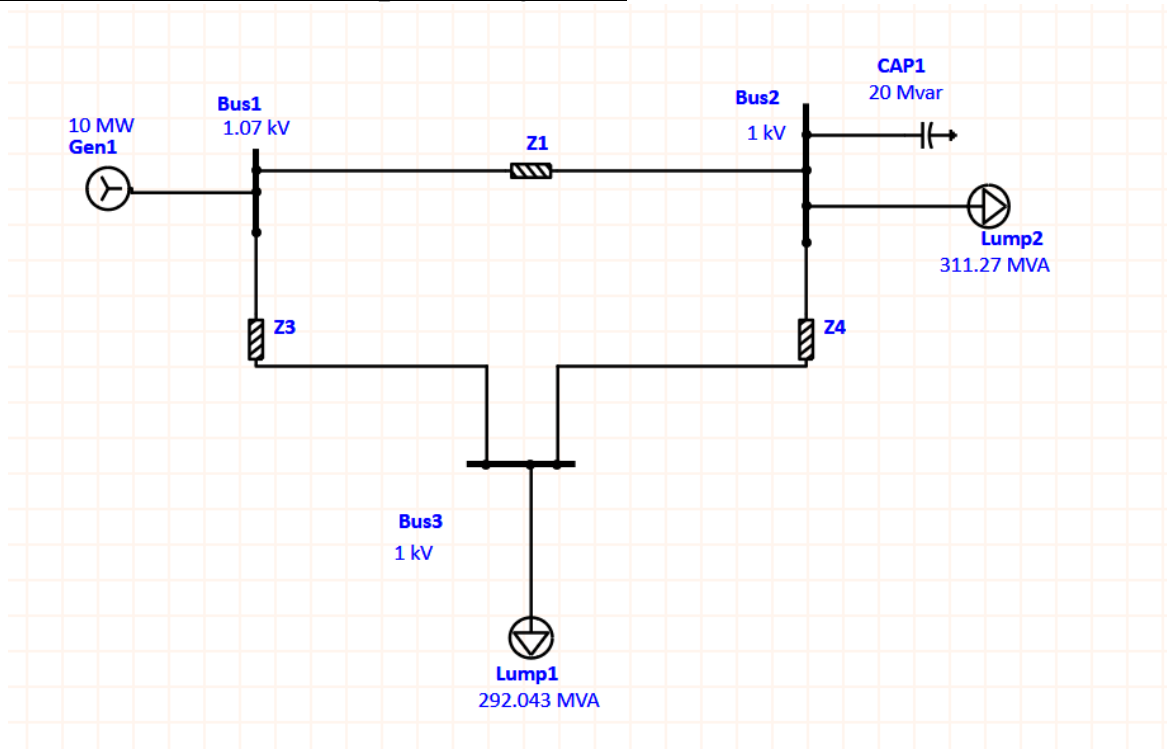
The Load Flow Analysis is a fundamental tool in interconnected power system analysis. Load Flow Analysis results play major role during the operational stages of any power system for its control and economic schedule, as well as during the expansion or design stages. The main purpose of any Load Flow Analysis is to compute precise steady-state voltages and voltage angles of all buses in the network, the reactive and real power flows into every overhead transmission line and power transformer (located at Sub-stations), under the assumption of known generation and load. Using a software such as ETAP eases the load flow calculations and analysis.



2 Power system model:

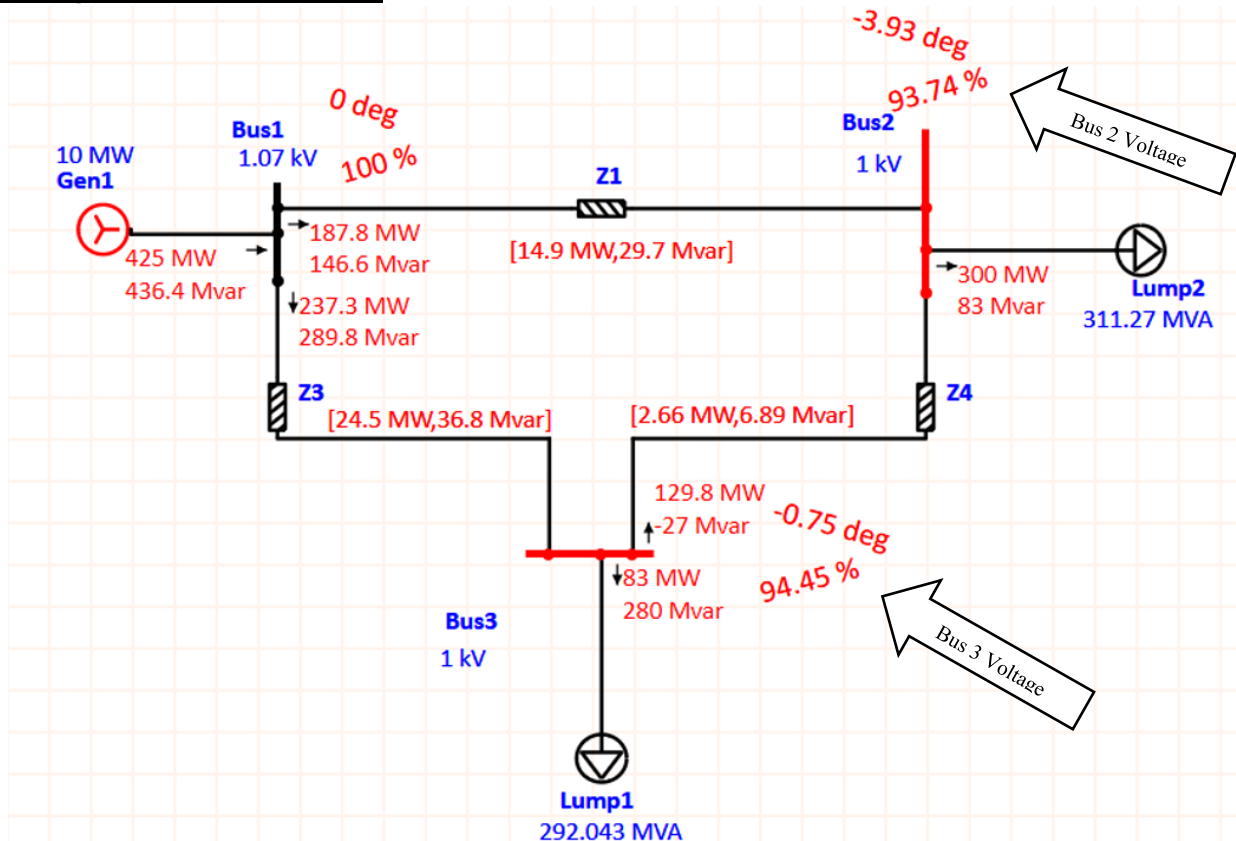


3 ETAP Simulation of the power system

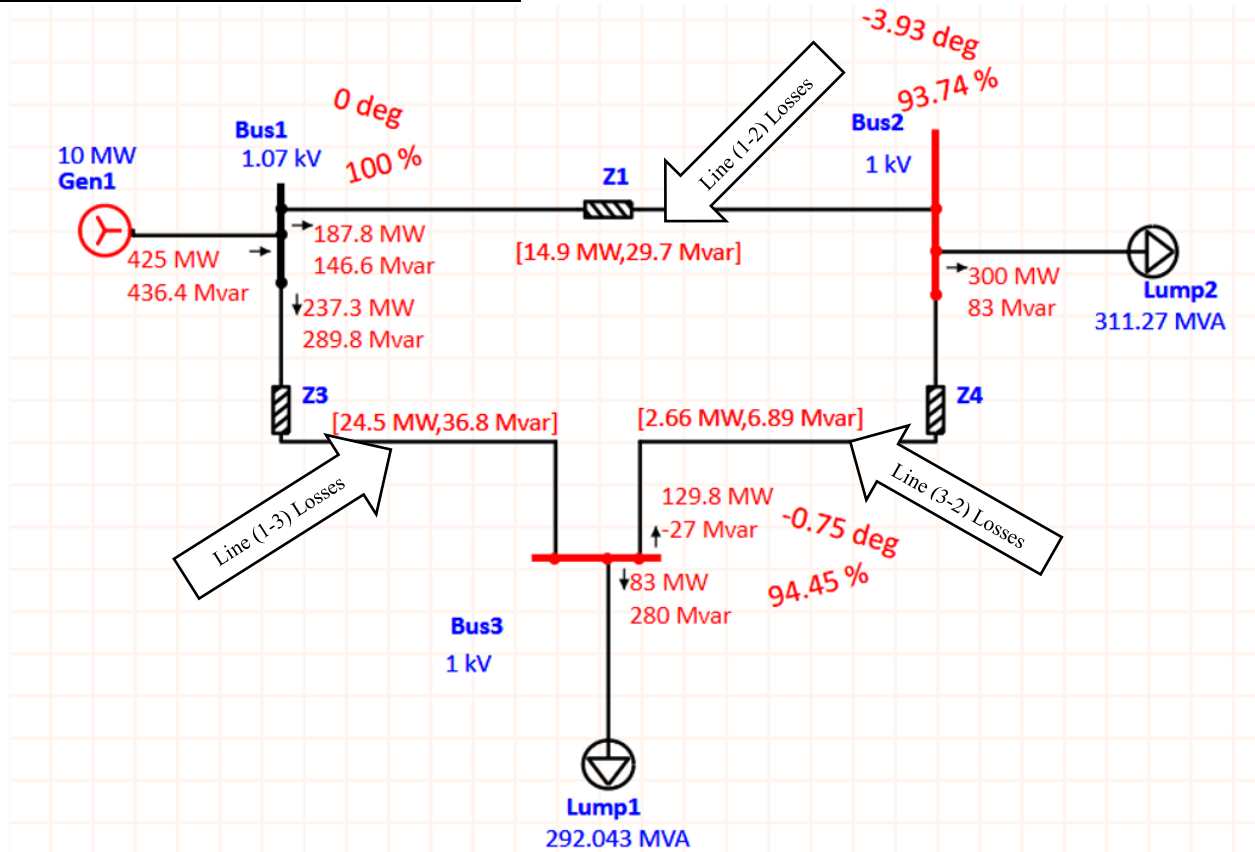


4 Snapshots of the ETAP

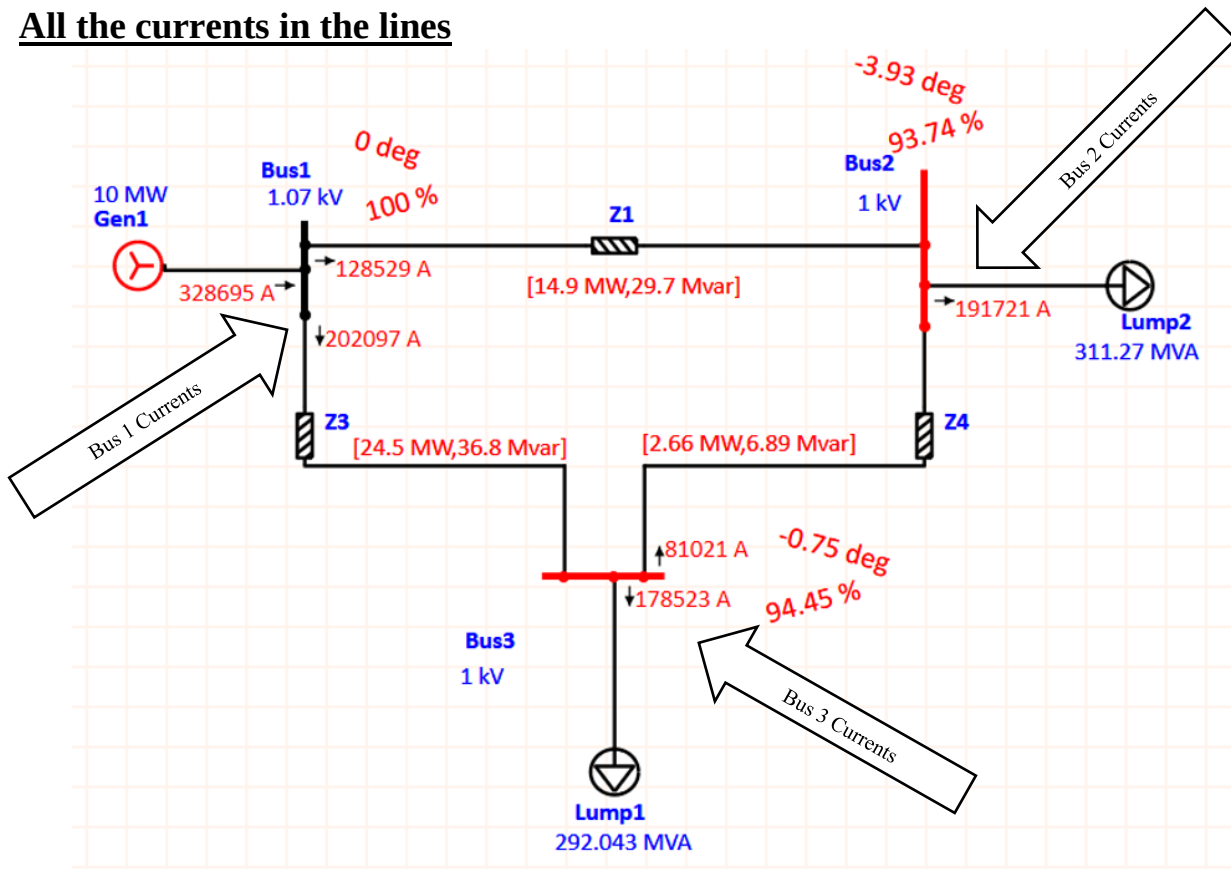
4.1 Voltages at buses 2 and 3



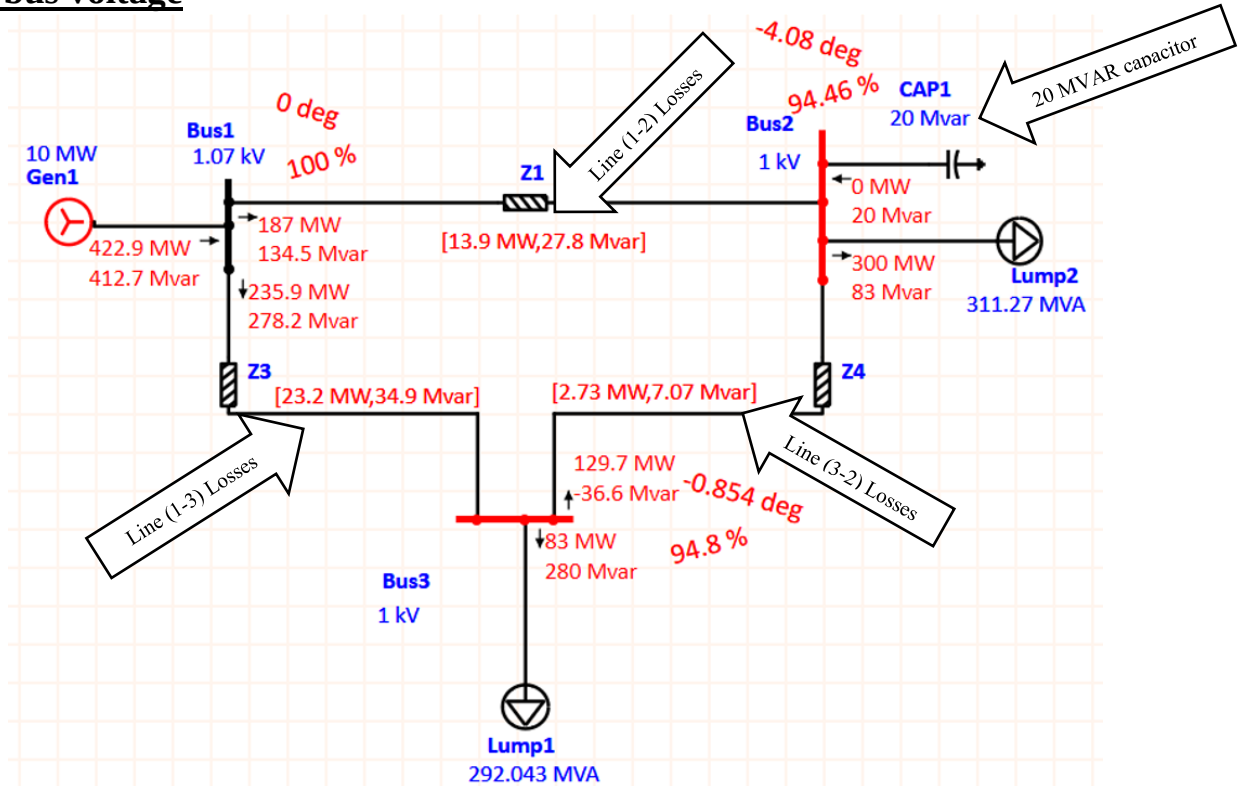
4.2 All active and reactive line losses.



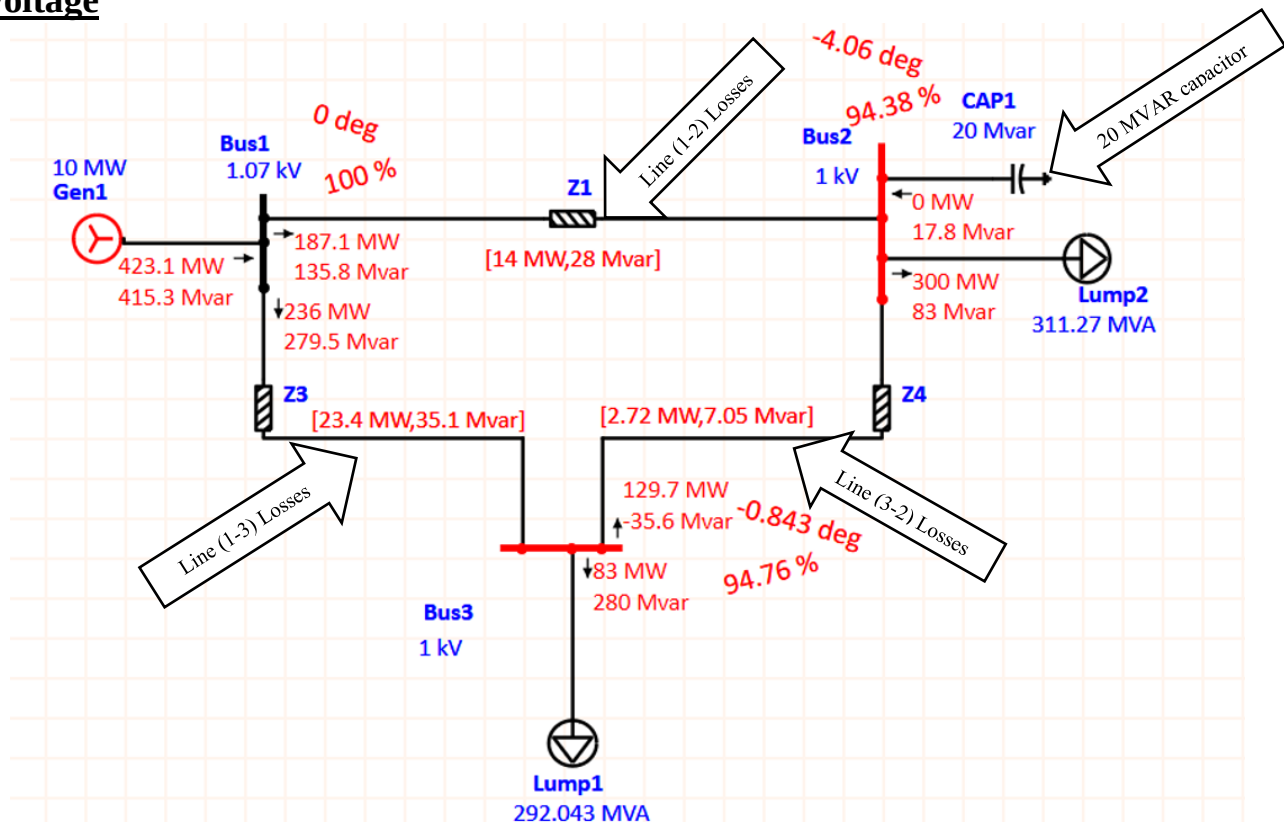
4.3 All the currents in the lines



4.4 The effect of 20 MVAR capacitor bank at bus 2 on the line losses, with voltage rating same bus voltage



4.5 The effect of 20 MVAR capacitor bank at bus 2 on the line losses, with 1kv rating bus voltage



4.6 The effect of the voltage magnitude and angle on the active and reactive power flow

From the simulation results obtained, it is easily shown that the active and reactive power flow from the higher voltage and angle (Ex: at bus (1) $1.07 < 0$) to the lower voltage and angle (at bus (2) $0.9374 < -3.93$). The following equations used to find the active and reactive power flow from each bus and how they are related to the bus voltage:

$$V_i^{[k+1]} = \frac{\frac{P_i^{[sch]} - jQ_i^{[sch]}}{V_i^{*[k]}} - \sum_{j=1, j \neq i}^n Y_{ij} V_j^{[k]}}{Y_{ii}}$$

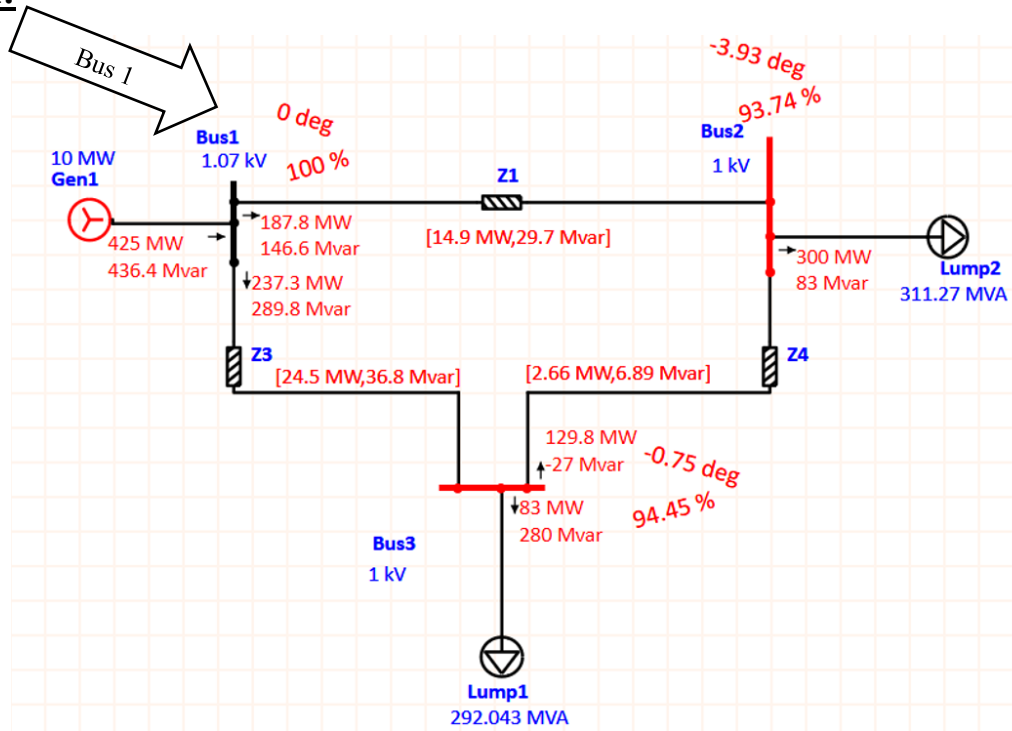
$$P_i^{[k+1]} = \Re \left\{ V_i^{*[k]} \left[V_i^{[k]} Y_{ii} + \sum_{j=1, j \neq i}^n Y_{ij} V_j^{[k]} \right] \right\}$$

$$Q_i^{[k+1]} = -\Im \left\{ V_i^{*[k]} \left[V_i^{[k]} Y_{ii} + \sum_{j=1, j \neq i}^n Y_{ij} V_j^{[k]} \right] \right\}$$

Case 1		Case 2	
Bus 1 Voltage and angle	$1.07 < 0$	Bus 1 Voltage and angle	$1.00 < 0$
Bus 2 Voltage and angle	$0.9374 < -3.93$	Bus 2 Voltage and angle	$0.8538 < -4.65$
Bus 3 Voltage and angle	$0.9445 < -0.75$	Bus 3 Voltage and angle	$0.8619 < -0.852$
Active power flow (1-2)	187.8 MW	Active power flow (1-2)	191.6 MW (↑)
Reactive power flow (1-2)	146.6 Mvar	Reactive power flow (1-2)	152.5 Mvar (↑)
Active power flow (1-3)	237.3 MW	Active power flow (1-3)	242.1 MW (↑)
Reactive power flow (1-3)	289.8 Mvar	Reactive power flow (1-3)	299.1 Mvar (↑)
Active power flow (2-3)	129.8 MW	Active power flow (2-3)	129.5 MW (↑)
Reactive power flow (2-3)	-27 Mvar	Reactive power flow (2-3)	-25.3 Mvar (↑)

In case we are comparing two different systems with changing same Bus Voltage, as we reduce the difference between Bus Voltages and the voltage angles, the active and reactive power flow increases. The following screenshots shows this concept.

4.6.1 Case 1:



4.6.2 Case 2:

